Testing the Dialect Mixture Hypothesis: Founding Population and Topolectal Variation in Quebec French

Technical handout — NWAV 2026, poster 284

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1. In brief

Question. Quebec French was built largely from linguistic features brought by settlers from many regions of France. Its pronunciation varies across the territory, mainly between a western area around Montreal and an eastern area around Quebec City. This variation is documented by the *Atlas linguistique de l'Est du Canada* (ALEC; Dulong & Bergeron 1980): phonetic transcriptions made in 1969–1973 at 169 survey points, from speakers born in the late 19th and early 20th centuries; we use the 125 localities in Quebec where Laurentian French is the historical variety. **Did the particular mix of French regional origins in each settlement hub determine this variation?** (For this view, see Rivard 1914; Poirier 1994: 74–75.)

Approach. We simulate ten 25-year generations (1620–1869) on the settlement network reconstructed by the *Programme de recherche en démographie historique* (PRDH). Two sources create differences between communities: the speech of the founding settlers, and local innovation. Three channels spread them: migration, contact between neighbouring parishes, and the spheres of influence of Quebec City and Montreal. The model predicts the phonetic distances between the 125 ALEC localities; its fit is measured by nested spatial cross-validation (12 regional blocks, each predicted by a model whose weights and tested values are chosen on the other 11 only).

Findings.

1. **No detectable gain from French regions.** Keeping every settler in the parish where they settled, we reassign their French birthplaces at random 100 times and refit the model. If regional origins shaped the map, the real birthplaces should fit better. They do not (R^2 0.77 vs. 0.75–0.80 for the reassigned ones), and the predicted maps are almost identical ($r \approx 0.99$).
2. **Founders, settlement history and contact explain most of the variation:** 78% of the estimated explainable variance (out-of-sample R^2 0.77; reliability ceiling 0.99). Local innovation adds almost nothing (estimated weight 2%).
3. **Interpretation.** The map is consistent with an early, slightly distinct norm in each of the first hubs (Quebec City, Montreal), whatever its exact source, spread through settlement and contact. This is compatible with new-dialect formation by mixing and levelling (Trudgill 2004), which the model does not simulate directly.

Imported ingredients, home-brewed map.

2. Data

2.1 Phonetic distances (ALEC)

For each entry of the atlas, we compute the Levenshtein distance between the phonetic transcriptions of every pair of localities, following Wieling & Nerbonne (2011), with substitution costs weighted by the difference in articulatory features between the two segments (insertions and deletions cost 0.5); distances are normalised by the length of the forms. The distance between two localities is the mean over the entries attested in both: 4,206 entries in all, each attested in a median of 37 of the 125 localities (only entries attested

in at least 17 survey points are kept). Rhotics ([r], [ʁ]) are kept distinct. Distances are small (0.009 to 0.037 between ALEC localities, mean 0.022) because most transcriptions are identical or nearly so from one locality to the next; what matters is how these small differences are distributed across the territory.

Fieldworkers differed in their transcription habits, which creates artificial boundaries between their survey areas (Saint-Amant Lamy 2026). The correction works in three steps:

1. For each fieldworker, distances between pairs of localities they both surveyed are shifted by a constant, so that neighbouring localities are on average as distant within the fieldworker's area as across its border.
2. Any remaining trend of these within-area distances with geographic distance, beyond the trend observed across areas, is removed.
3. The corrected distances are smoothed over neighbouring localities: each distance is averaged, with weight 0.5, with those of the localities' neighbours in a Delaunay triangulation.

These corrections and the smoothing are applied once, to the full matrix, before any model fitting; the cross-validation of section 4 therefore does not evaluate them, and the R^2 is not a fit to raw, unprocessed transcriptions.

Reliability ceiling. Entries are split at random into two halves, each corrected and smoothed separately. The correlation between the two halves, extended to the full set of entries by the Spearman–Brown correction (20 random splits), yields an estimated reliability of 0.99, which we use as an approximate ceiling for predictive R^2.

2.2 Settlement (PRDH)

The PRDH reconstructs, from parish registers, the people who settled the St. Lawrence Valley and its hinterland (Dillon et al. 2018). From it we trace how the territory was occupied, generation by generation (ten generations of 25 years, 1620–1869): which localities existed, how many people lived there, and where newcomers came from. Movements are measured from complete life trajectories in parish records (baptism → first marriage, marriage → burial, and baptism → burial for unmarried individuals who died at 15 or older), across 434 localities. A new parish starts with the mixture of its founders.

Marriage records end in 1824. For the last two generations, movements are inferred from trajectories ending in a burial (recorded until 1861): from the place of baptism or marriage to the place of burial. Flows between existing parishes are kept at their last fully observed level (1795–1819); parishes founded after 1824 draw their founders from the baptism or marriage places of the people buried there.

Each ALEC locality is linked to the PRDH locality (or nearby group of localities) it descends from. These associations were checked one by one.

2.3 French birthplaces

The immigrants' birthplaces come from the *Fichier Origine* of the Fédération québécoise des sociétés de généalogie. A **pioneer** is an immigrant with a known, geolocated birthplace in France, who arrived before 1763 and settled in a parish during its first 50 years. This gives 3,796 pioneers born in 2,394 French communes.

3. Model

3.1 State

For each generation *t* (25 years), the model tracks a matrix $\mathbf{S}_t$ over the PRDH localities: $S_t(i, j)$ measures how much the speech inputs of localities *i* and *j* have in common. Each generation, the state is passed on through the channels, and the sources add new differences:

$$\mathbf{S}_t = \mathbf{A}_t \mathbf{S}_{t-1} \mathbf{A}_t^\top + \mathbf{E}_t, \text{ with } \mathbf{A}_t = \mathbf{Z}_t \mathbf{G}_t \mathbf{M}_t$$

where $\mathbf{M}_t$ is migration, $\mathbf{G}_t$ neighbour contact, $\mathbf{Z}_t$ the spheres of influence, and $\mathbf{E}_t$ the new inputs. Each row of $\mathbf{A}_t$ gives the share of a locality's speech inherited from each locality at the previous generation. Localities not yet founded are left out. The two sources are propagated separately, giving $\mathbf{S}^{\text{settlers}}$ and $\mathbf{S}^{\text{innov}}$ in 1869. **S** is a covariance matrix: its diagonal measures how much each locality's inputs vary, its off-diagonal terms how much two localities share. It is converted into expected differences in section 3.4.

3.2 Sources

Founding settlers. Each pioneer *p* adds the speech patterns of their birthplace to the parish where they settled. Pioneers' speech is not measured: the model assumes that two pioneers born *d* km apart spoke alike in proportion to $k(d) = 1 / (1 + d / 100 \text{ km})$ (Figure 1), so that similarity halves at 100 km and keeps declining. The choice of 100 km matters little: with 25 km or 400 km, the expected differences between ALEC localities correlate above 0.998 with those at 100 km. A pioneer's contribution is divided by the parish's mean population over its first 50 years, so that a pioneer counts more in a small parish. Only pioneers arriving before 1763, in the first 50 years of their parish, contribute. This is an abstract mixture: the model represents how much of each birthplace's speech patterns a parish inherits, not individual linguistic features, and it does not simulate competition or majority selection between variants (Trudgill 2004); no French dialect data are used.

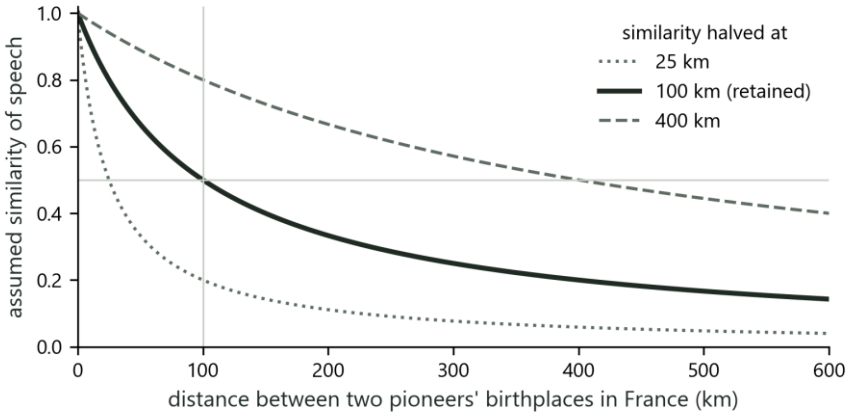


Figure 1. Assumed similarity of two pioneers' speech as a function of the distance between their birthplaces in France.

Local innovation. Each generation, every existing parish receives an independent new input of the same size, so that parishes drift apart. Notre-Dame-de-Québec receives ten times more from 1620 to 1763 (plausible range 3–30; the data cannot distinguish values within this range). This component deliberately groups changes arising within the colony and later inputs from outside (e.g. later immigrants, prestige norms); it is not split into endogenous and exogenous parts.

3.3 Channels

Migration (M_t), fixed. The share of each locality's population coming from each other locality during the generation (section 2.2).

Neighbour contact (G_t), a gravity model (Trudgill 1974). Locality i takes up a share r_i of its speech patterns from other localities, distributed in proportion to $P_j \cdot \exp(-d_{ij} / s)$ (P_j: population; the exponent on population is fixed at 1). A parish with fewer or smaller neighbours takes up less: $r_i = r \cdot \min(1, N_i / N_{\text{ref}})$, where $N_i = \Sigma_j P_j \exp(-d_{ij} / s)$, and N_{ref} is the value for a typical riverside parish with two neighbours 7 km away. Tested: $r \in \{0, 0.05, 0.10, 0.20\}$ per generation, $s \in \{5, 10, 25\}$ km.

Spheres of influence (Z_t). Each locality belongs to the city, Quebec City or Montreal, that attracts it more, with attraction $\log P_c - d_{ic} / s_z$ (P_c: population of the urban area). It takes up a share $r_z \cdot \exp(-d_{ic} / s_z)$ of its speech patterns from the city's main parish. The cities themselves take up nothing. Tested: $r_z \in \{0, 0.05, 0.10, 0.20\}$, $s_z \in \{50, 100, 200\}$ km.

Why these ranges. A generation lasts 25 years, and migration alone already renews about half of a parish's population (median, from the PRDH flows). Contact without moving is a weaker force: we capped each channel at 20% of a parish's speech per generation, beyond which contact would quickly erase the differences that settlement creates, whereas regional differences are still recorded two centuries later. Reaches follow the scale of each kind of contact: 5 to 25 km for neighbour contact, about the distance between neighbouring parishes (typically 5 to 10 km apart, median 7 km) and within a day's travel for markets, marriages or church; 50 to 200 km for the spheres of influence, the scale of a city's hinterland for trade, administration and the clergy (Trois-Rivières lies 115 km from Quebec City and 122 km from Montreal). All ranges were set before fitting the model to the ALEC, and simulations confirmed that, within them, the model can tell the channels and their values apart.

3.4 From the model to the ALEC

Each ALEC locality is a weighted combination of the PRDH localities it descends from (section 2.2). For each source, the expected difference between ALEC localities i and j is $C_{ij} = S_{ii} + S_{jj} - 2S_{ij}$, scaled to a mean of 1. The observed distance is modelled as

$$D_{ij} \approx \alpha + \sqrt{c + w_{\text{settlers}} C_{ij}^{\text{settlers}} + w_{\text{innov}} C_{ij}^{\text{innov}}}, \quad w \geq 0, \qquad c \geq 0$$

and smoothed like the observed distances (section 2.1). The parameters are estimated by weighted least squares (weights: number of entries shared by the pair).

The parameters. *α (baseline)* is the distance predicted for two localities with identical speech inputs. It captures what all pairs share and the model does not explain: individual variation among speakers, and transcription differences left after correction. No pair of ALEC localities is closer than about 0.009. *w (weight of a source)* converts the model's expected differences, which have no unit, into phonetic distance. Because each source's expected differences are scaled to a mean of 1, the weights of different sources can be compared directly: they give each source's share of the predicted variation, within the model; they are not the share of the observed variance attributable to a historical cause. *c (noise)* adds the same amount to the expected difference of every pair. It absorbs variation that is specific to each locality and shared with none of its neighbours; here it is small (0.0000037), because smoothing has already removed most of it. All three are estimated from the data, *w* and *c* constrained to be non-negative.

Why a square root. *C* is an expected squared difference, like a variance; observed distances behave rather like a standard deviation. Taking the square root converts expected squared differences into a distance-like quantity, on a scale more comparable to the observed phonetic distances.

Worked example: Acton Vale and Notre-Dame-de-Ham

Two ALEC localities 71 km apart, on either side of the east–west transition. Retained model: founding settlers, migration, neighbour contact (20%, 25 km) and spheres of influence (20%, 200 km). Parameters fitted on all pairs, for illustration (see 3.4): $\alpha = 0.0125$, $c = 0.0000037$, $w = 0.000111$.

Step 1. Shared speech inputs in 1869 (founding settlers, scaled so that the mean expected difference is 1): $S_{ii} = 70.47$ (Acton Vale), $S_{jj} = 86.80$ (Notre-Dame-de-Ham), $S_{ij} = 77.87$. These values are large because all parishes share a common colonial base; only differences matter.

Step 2. Expected difference. $C_{ij} = S_{ii} + S_{jj} - 2\,S_{ij} = 70.47 + 86.80 - 2 \times 77.87 = 1.53$, about one and a half times the average difference between two ALEC localities.

Step 3. From difference to distance. $\sqrt{(c + w \cdot C_{ij})} = \sqrt{(0.0000037 + 0.000111 \times 1.534)} = 0.0132$; $\alpha + 0.0132 = 0.0256$.

Step 4. Smoothing, as for the observed distances. The value is averaged with those of each locality's neighbours. Acton Vale's five neighbours (Saint-Hyacinthe, Granby, Saint-Guillaume, L'Avenir, Saint-Étienne-de-Bolton) average 0.0120 from Notre-Dame-de-Ham; Notre-Dame-de-Ham's seven neighbours average 0.0097 from Acton Vale. $0.5 \times 0.0132 + 0.25 \times (0.0120 + 0.0097) = 0.0120$; $\alpha + 0.0120 = 0.0245$.

Result. Predicted distance 0.0245; observed 0.0266. For comparison, Saint-Chrysostome and Beloeil, at about the same distance (68 km) but on the same side of the transition, descend largely from the same founders and share a sphere of influence: $C = 0.001$, predicted 0.015, observed 0.010.

4. Evaluation

4.1 Nested spatial cross-validation

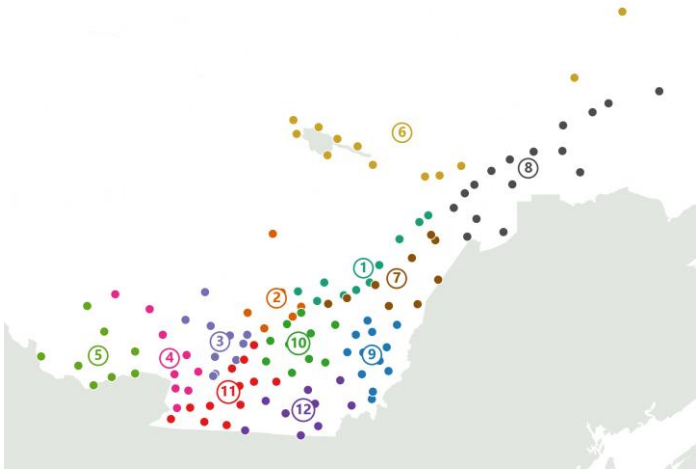


Figure 2. The 12 cross-validation blocks. North shore: 1 Quebec City area, 2 Trois-Rivières area, 3 Montreal north shore–Lanaudière, 4 Laurentides–Vaudreuil, 5 Outaouais, 6 Saguenay–Lac-Saint-Jean–Côte-Nord. South shore: 7 Côte-du-Sud, 8 Bas-Saint-Laurent–Gaspésie, 9 Beauce–Appalaches, 10 Nicolet–Bois-Francs, 11 Richelieu–Suroît, 12 Eastern Townships.

The 125 ALEC localities are grouped into 12 regional blocks, six on each shore of the St. Lawrence (Figure 2). Each block is held out in turn:

1. Localities within 25 km of the held-out block, on the same shore, are set aside as a buffer, so that close neighbours do not inform the prediction. This matters all the more because smoothing (section 2.1)

averages each distance with those of neighbouring localities: without a buffer, information from the held-out block would leak into the training pairs.

2. On the remaining localities, the tested values of the channels (share and reach) are chosen by an inner cross-validation over the other blocks.
3. With these values, α, w and c are estimated on the remaining localities.
4. The model then predicts every pair involving a held-out locality.

The out-of-sample R^2 compares these predictions with a constant (the mean training distance), pooled over the 12 blocks. The selection of tested values is thus part of what is evaluated: the R^2 is an out-of-sample estimate for the whole procedure (selection and fitting). Two things are not nested: the preprocessing of the ALEC matrix (section 2.1), and the choice of the model's structure (which sources and channels), made by comparing the combinations of section 4.2 and over the course of the project. The latter is why the model is described as exploratory.

4.2 All combinations of sources and channels

Every combination was fitted with the same procedure, each choosing its own channel values (out-of-sample R^2):

Sources \ channels	settlement only	+ neighbour contact	+ spheres of influence	+ both
Founding settlers	0.43	0.65	0.70	**0.77**
Local innovation	0.13	0.24	0.41	0.45
Both sources	0.43	0.65	0.70	0.77

Founding settlers carry the structure; local innovation adds almost nothing once settlers are in the model (with both sources and both channels, founding settlers account for 98% of the fitted source contributions, and local innovation for 2%). Each channel improves the fit, and the two together do best.

4.3 Retained model

Founding settlers, migration, neighbour contact and spheres of influence: R^2 0.77, that is 78% of the estimated explainable variance (reliability ceiling 0.99). Adding local innovation as a second source gives the same R^2 (0.77; estimated weight 2%), so the simpler model is retained; it is the one used in the shuffle test (section 5). All 12 blocks choose the same channel values: neighbour contact 20% per generation within 25 km, spheres of influence 20% within 200 km. These are the upper ends of the tested ranges (section 6). Parameters fitted on all pairs (for illustration): $\alpha = 0.0125$, $w = 0.000111$, $c = 0.0000037$.

4.4 Sensitivity to the design of the channels

R^2 of the retained model, refitted with the same procedure:

Variant	R^2
Retained model	0.773
Neighbour contact weighted by √population	0.764
Neighbour contact by distance only (no population)	0.755
Trois-Rivières as a third city	0.775

The same channel values are chosen in every variant.

5. Testing French origins: the shuffle test

5.1 Design

If the regional origins of the founders shaped the map, the model should fit better with the settlers' real birthplaces than with birthplaces that have lost their geography. We build 100 such alternatives, keeping every pioneer in the parish where they settled:

- **50 permutations:** the birthplaces are randomly reassigned among the pioneers. The same set of birthplaces is used, but who was born where is scrambled.
- **50 relabellings:** the French communes are randomly swapped with one another. Pioneers born in the same commune still share a birthplace, but the communes' positions in France are scrambled.

For each shuffle, the retained channel values are held fixed, while α, w and c are re-estimated within each outer cross-validation fold, on the training pairs only and with the same buffers as for the retained model.

5.2 Result

Birthplaces	R^2 (real) / median R^2 (shuffles)	Range	Real birthplaces better than
Real	0.773		
Permuted among settlers (50)	0.786	0.759–0.803	2 of 50
Communes relabelled (50)	0.777	0.751–0.803	15 of 50

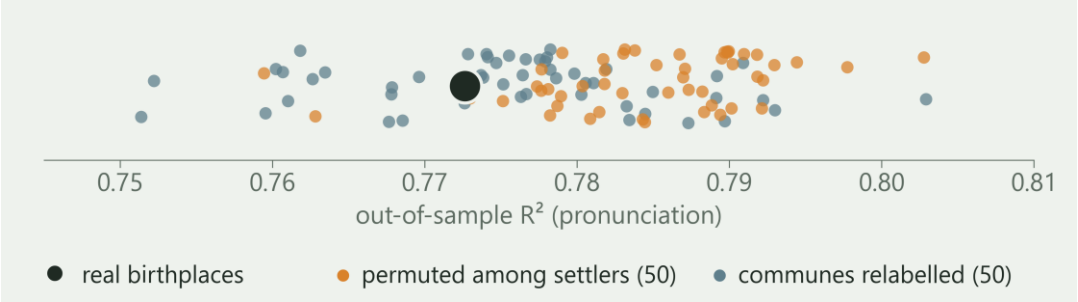


Figure 3. Out-of-sample R^2 with the real birthplaces and with 100 shuffles.

Shuffling changes the predicted map very little: the expected differences between localities correlate at about 0.99 with and without shuffling. Relabellings, which keep everything about the pioneers' origins except where the communes lie in France, are the most direct test of geography: the real birthplaces fall in their lower half. Permutations, which also separate pioneers born in the same commune, fit slightly better still. The results suggest that the predicted map depends much more on early differences between the first settlement hubs (Quebec City, Montreal), whatever their source, and their subsequent propagation through settlement and contact, than on the precise French geography of the founders. Differences arising everywhere over time do not produce this map (local innovation alone: R^2 0.45, section 4.2). In this model, the founders' French regions bring no detectable predictive gain.

Our interpretation: each hub quickly formed its own slightly distinct variety by selecting among the many speech patterns its pioneers brought. Their French origins shaped the pool of available patterns, but do not detectably predict which ones prevailed in each hub.

5.3 What the test can show

On simulated data in which the truth is known, real birthplaces tend to rank high among the shuffles when regional origins matter, and low when they do not. The ALEC result is closer to the second case. Because real and shuffled birthplaces predict nearly the same map, however, the test cannot rule out a weak regional effect. The test establishes that, in this model, the real French locations of the birthplaces do not improve prediction. It does not establish that regional origins had no historical effect: a weak effect cannot be ruled out, and the power of the test has not been quantified.

6. Limits and open questions

A. Limits bearing on the test of French origins

- **No French dialect data.** The model does not use the speech of the French regions; it assumes that pioneers born close together spoke alike. Similarity is approximated by distance between birthplaces, without explicit historical dialect boundaries.
- **What the shuffle test can show.** Real and shuffled birthplaces predict nearly the same map. The test shows that regional origins bring no detectable gain; it cannot rule out a weak regional effect, and the power of the test has not been quantified.
- **Linear transmission.** Each parish's speech is modelled as a weighted mixture of its founders' inputs. If a majority variant tended to win outright (Trudgill 2004), small differences in the founding mix could have larger effects than the model allows.

B. Limits bearing on predictive performance

- **Channels at the upper end of their ranges.** Both channels choose the strongest share (20% per generation) and the longest reach tested. Contact may therefore have been stronger, or reached farther, than the ranges allow. The shuffle test compares the real and reassigned birthplaces at the same retained channel values; its sensitivity to values beyond the tested ranges remains unknown.
- **Preprocessing.** The fieldworker correction and the smoothing are applied once, before fitting (section 2.1); the R^2 is not a fit to raw transcriptions.
- **The last generations.** The model stops in 1869; the ALEC speakers were born in the late 19th and early 20th centuries. Urbanisation and emigration (to the cities and New England) in between are not modelled. Settlement after 1824 also rests on incomplete data (section 2.2).
- **What remains unexplained.** The model accounts for 78% of the estimated explainable variance. The remaining 22% includes more local patterns, some of which may reflect transcription habits not fully removed by the correction.

C. Limits bearing on the historical interpretation

- **Exploratory status.** The model was refined over the course of the project, after earlier versions had been compared with the ALEC; the findings should be read as exploratory. The choice of the model's structure was not nested in the cross-validation (section 4.1).
- **A plausible model, not a unique causal chain.** The model shows that this history can produce the observed map; it does not identify a unique historical sequence. The early formation of distinct norms in Quebec City and Montreal is an interpretation compatible with the maps, not a fact observed in the 17th century.
- **Local innovation as one component.** Changes arising within the colony and later inputs from outside are deliberately grouped (section 3.2).
- **Display only.** On the poster's settlement map, one long-distance founding arrow (Charlevoix → Outaouais, 1845–1869) is omitted for legibility; the model uses it.

7. References, data and acknowledgements

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Data sources

ALEC: published atlas (Dulong & Bergeron 1980); transcriptions digitised by two teams, Yves Charles Morin and Luc Baronian, then Hugo Saint-Amant Lamy and the CRIFUQ. Settlement: Programme de recherche en démographie historique (PRDH), Université de Montréal. French birthplaces: *Fichier Origine*, Fédération québécoise des sociétés de généalogie. Basemaps: Natural Earth (public domain).

Code and data availability

The analysis code and derived data are available from the author on request. This handout and the poster are available at github.com/hugosaintamantlamy/NWAV_2026.

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